

COMPLETE 2-DESCENT: AN EXAMPLE

PREVIOUSLY...

Weil pairing + Kummer pairing + Hilbert's Thm 90

→ Bilinear pairing $b: E(K)/_m E(K) \times E[m] \rightarrow K^x / (K^x)^m$
s.t. $e_m(K(P, \sigma), T) = \delta_K(b(P, T))(\sigma)$.

→ induces:
$$\begin{array}{ccc} E(K)/_m E(K) & \hookrightarrow & \text{Hom}(E[m], K^x / (K^x)^m) \\ P & \mapsto & b(P, \cdot) \end{array}$$

s.t. $b(P, T) \in K(S, m) = \{ b \in K^x / (K^x)^m : \text{ord}_v(b) \equiv 0 \pmod m \ \forall v \notin S \}$

w/ $S = M_K^\infty \cup \{ \text{bad primes} \} \cup \{ \text{primes} \mid m \} \subseteq M_K$

and $b(P, T) \equiv f_T(P) \pmod{(K^x)^m}$.

Thm (Complete 2-descent)

$$E/K : y^2 = (x-e_1)(x-e_2)(x-e_3) \quad e_i \in K, e_i \neq e_j.$$

Let $S = M_K^\infty \cup \{ \text{bad places of } E/K \} \cup \{ \text{places dividing } 2 \}$

Let $K(S, 2) = \{ b \in K^\times / (K^\times)^2 : \text{ord}_v(b) \equiv 0 \pmod{2} \quad \forall v \notin S \}$

Then, there is an injective homomorphism: $E(K) / 2E(K) \rightarrow K(S, 2) \times K(S, 2)$

defined by

$$P = (x, y) \mapsto \begin{cases} (x-e_1, x-e_2) & \text{if } x \neq e_1, e_2 \\ \left(\frac{e_1-e_3}{e_1-e_2}, e_1-e_2 \right) & \text{if } x = e_1 \\ \left(e_2-e_1, \frac{e_2-e_3}{e_2-e_1} \right) & \text{if } x = e_2 \\ (1, 1) & \text{if } P = \mathcal{O}. \end{cases}$$

Moreover:

If $(b_1, b_2) \in K(S, 2)^2$ is the image of $P \neq \mathcal{O}, T_1, T_2$ then

$$P = (x, y) \iff \begin{cases} b_1 z_1^2 - b_2 z_2^2 = e_2 - e_1 \\ b_1 z_1^2 - b_1 b_2 z_3^2 = e_3 - e_1 \end{cases} \quad \text{for } (z_1, z_2, z_3) \in K^\times \times K^\times \times K$$

and $P = (b_1 z_1^2 + e_1, b_1 b_2 z_1 z_2 z_3)$.

EXAMPLE (Find $E(\mathbb{Q})$)

Let $E/\mathbb{Q} : y^2 = x^3 - 12x^2 + 20x = x(x-2)(x-10)$

$$\Delta_E = 409600 = 2^{14} \cdot 5^2, \quad C_4 = 2^6 \cdot 3 \cdot 7.$$

There is bad reduction at 2 (additive), 5 (multiplicative).

Torsion: $E[2] \subseteq E(\mathbb{Q})$, more? Recall: $E(\mathbb{Q})[m] \hookrightarrow \tilde{E}(\mathbb{F}_p)$ P is good
PTm

$p=3$, $\# \tilde{E}(\mathbb{F}_3) = 4$
(good) no prime-to-3 torsion
other than $E[2]$

$p=11$, $\# \tilde{E}(\mathbb{F}_{11}) = 16 \Rightarrow E(\mathbb{Q})_{\text{tors}} = E(\mathbb{Q})[2]$.
(good) no 3-torsion.

$$S = \{\infty, 2, 5\}, \quad \mathcal{Q}(S, 2) = \left\{ b \in \frac{\mathbb{Q}^\times}{(\mathbb{Q}^\times)^2} : \text{ord}_p(b) \equiv 0 \pmod{2} \forall p \notin S \right\}$$

Thus,
$$= \{ \pm 1, \pm 2, \pm 5, \pm 10 \} \quad (\text{size } 8)$$

$$\frac{E(\mathbb{Q})}{2E(\mathbb{Q})} \cong \underbrace{\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus (\mathbb{Z}/2\mathbb{Z})}_{\text{size } 2^{R+2}} \xrightarrow{R \in \mathbb{Q}} \underbrace{\mathcal{Q}(S, 2) \times \mathcal{Q}(S, 2)}_{\text{size } 8^2 = 2^6} \left. \vphantom{\begin{matrix} \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus (\mathbb{Z}/2\mathbb{Z}) \\ \mathcal{Q}(S, 2) \times \mathcal{Q}(S, 2) \end{matrix}} \right\} \begin{matrix} R+2 \leq 6 \\ \rightarrow R \leq 4 \end{matrix}$$

$$(c) \begin{cases} b_1 z_1^2 - b_2 z_2^2 = 2 \\ b_1 z_1^2 - b_1 b_2 z_3^2 = 10 \end{cases}$$

$5 \nmid b_1$ and $5 \mid b_2$ same proof!

Case $(b_1, b_2) = (1, 5)$

$$(c) \begin{cases} (1) z_1^2 - 5 z_2^2 = 2 \\ (2) z_1^2 - 5 z_3^2 = 10 \end{cases}$$

$$\mathbb{Q} \subseteq \mathbb{Q}_5$$

Work over $\mathbb{Q}_5$:

$$\nu_p(xy) = \nu_p(x) + \nu_p(y)$$

$$\nu_p(x+y) \geq \min \{ \nu_p(x), \nu_p(y) \}$$

↑ equal if $\nu_p(x) \neq \nu_p(y)$

Claim 2 $z_3 \in \mathbb{Z}_5$

$$(2) \nu_5(z_1^2 - 5 z_3^2) = \nu_5(10) = 1$$

$$\min \{ 2\nu_5(z_1), 1 + 2\nu_5(z_3) \} \Rightarrow \nu_5(z_3) \geq 0$$

Claim 1 $z_1, z_2 \in \mathbb{Z}_5$

$$(1) \nu_5(z_1^2 - 5 z_2^2) = \nu_5(2) = 0$$

$$\min \{ 2\nu_5(z_1), 1 + 2\nu_5(z_2) \}$$

$$\Rightarrow \nu_5(z_1), \nu_5(z_2) \geq 0. \rightarrow z_1, z_2 \in \mathbb{Z}_5.$$

From (1), $z_1, z_2 \in \mathbb{Z}_5 \Rightarrow z_1^2 \equiv 2 \pmod{5}$ but $2 \not\equiv \square \pmod{5}$ $\therefore$
 $\Rightarrow$ No $\mathbb{Q}_5$ solutions!

No $\mathbb{Q}$ -soln's!!

$$(b_1, b_2) = (1, -1) \quad \begin{cases} z_1^2 + z_2^2 = 2 \\ z_1^2 + z_3^2 = 10 \end{cases}$$

$$(z_1, z_2, z_3) = (1, 1, 3)$$

Conclusion

$$E(\mathbb{Q}) / 2E(\mathbb{Q}) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/2$$

$$\Rightarrow E(\mathbb{Q}) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}$$

" and $\text{rank } E(\mathbb{Q}) = 1$.

$$\langle (0, 0), (2, 0), (1, -3) \rangle$$

rep'n
 $p = (1, -3)$

SELMER AND SHA

Group cohomology: $0 \rightarrow P \rightarrow M \rightarrow N \rightarrow 0$ G_k -modules

$$\Rightarrow 0 \rightarrow H^0(G_k, P) \rightarrow H^0(G_k, M) \rightarrow H^0(G_k, N)$$

$$\xrightarrow{\delta} H^1(G_k, P) \rightarrow H^1(G_k, M) \rightarrow H^1(G_k, N)$$

Consider: $0 \rightarrow E[m] \rightarrow E \xrightarrow{[m]} E \rightarrow 0$ ← $E(\bar{k})$

$$\Rightarrow 0 \rightarrow E(k)[m] \rightarrow E(k) \xrightarrow{[m]} E(k) \xrightarrow{\delta} H^1(G_k, E[m]) \rightarrow H^1(G_k, E) \xrightarrow{[m]} H^1(G_k, E)$$

Extract: $0 \rightarrow \frac{E(k)}{mE(k)} \xrightarrow{\delta} H^1(G_k, E[m]) \rightarrow H^1(G_k, E)[m] \rightarrow 0$

$$P \mapsto \phi : G_k \rightarrow E[m] \quad \text{where } [m]Q = P.$$

$$\sigma \mapsto Q^\sigma - Q$$

If $E[m] \subseteq E(k)$, $\frac{E(k)}{mE(k)} \hookrightarrow \text{Hom}(G_k, E[m]) \quad P \mapsto k(P, \cdot)$

More generally: suppose E, E' are isogenous over K
 $\phi: E \rightarrow E'/K$, $\phi \circ \hat{\phi} = [m]$.

We have: $0 \rightarrow E[\phi] \rightarrow E \xrightarrow{\phi} E' \rightarrow 0$

Cohom: $0 \rightarrow E(K)[\phi] \rightarrow E(K) \xrightarrow{\phi} E'(K) \xrightarrow{\delta} H^1(G_K, E[\phi]) \rightarrow H^1(G_K, E) \xrightarrow{\phi} H^1(G_K, E)$

Extract: $0 \rightarrow \frac{E'(K)}{\phi(E(K))} \xrightarrow{\delta} H^1(G_K, E[\phi]) \rightarrow H^1(G_K, E)[\phi] \rightarrow 0$

Also locally! Fix $\nu \in M_K$, $K \subseteq K_\nu$, $\bar{K} \subseteq \bar{K}_\nu$, $G_\nu = \text{Gal}(\bar{K}_\nu/K_\nu)$

then $G_\nu \subseteq \text{Gal}(\bar{K}/K)$ as a decomposition gp at ν .

Also $E(\bar{K}) \subseteq E(\bar{K}_\nu)$ and G_ν acts on $E(\bar{K}_\nu)$

Obtain: $0 \rightarrow \frac{E'(K_\nu)}{\phi(E(K_\nu))} \rightarrow H^1(G_\nu, E[\phi]) \rightarrow H^1(G_\nu, E)[\phi] \rightarrow 0$

$G_\nu \subseteq G_K$, $E(\bar{K}) \subseteq E(\bar{K}_\nu)$ gives restriction maps in cohom.

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \frac{E'(k)}{\phi(E(k))} & \xrightarrow{f} & H^1(G_k, E[\phi]) & \longrightarrow & H^1(G_k, E)[\phi] \longrightarrow 0 \\
 & & \downarrow & \curvearrowright & \downarrow & \curvearrowright & \downarrow \\
 0 & \longrightarrow & \prod_{v \in M_k} \frac{E'(k_v)}{\phi(E(k_v))} & \longrightarrow & \prod_{v \in M_k} H^1(G_v, E[\phi]) & \longrightarrow & \prod_{v \in M_k} H^1(G_v, E)[\phi] \longrightarrow 0
 \end{array}$$

Goal: $\text{Image}(f) = \text{kernel}(H^1(G_k, E[\phi]) \rightarrow H^1(G_k, E)[\phi])$
 $\subseteq \text{kernel}(H^1(G_k, E[\phi]) \rightarrow \prod H^1(G_v, E))$

Def. ϕ -Selmer gp of E/k is the subgroup of $H^1(G_k, E[\phi])$

$$S^{(\phi)}(E/k) = \text{Ker} \{ H^1(G_k, E[\phi]) \rightarrow \prod_{v \in M_k} H^1(G_v, E) \}$$

$$\underline{III}(E/k) = \text{Ker} \{ H^1(G_k, E) \rightarrow \prod_{v \in M_k} H^1(G_v, E) \} \quad \text{Tate-Shafarevich gp}$$

